Pixels and Processing

YJ40201: Fundamentals of Programming

Stony Brook Institute at Anhui University

Outline of Topics

- Pixels and Coordinate Systems
- Drawing Simple Shapes
- Introduction to the Processing Environment
- Color

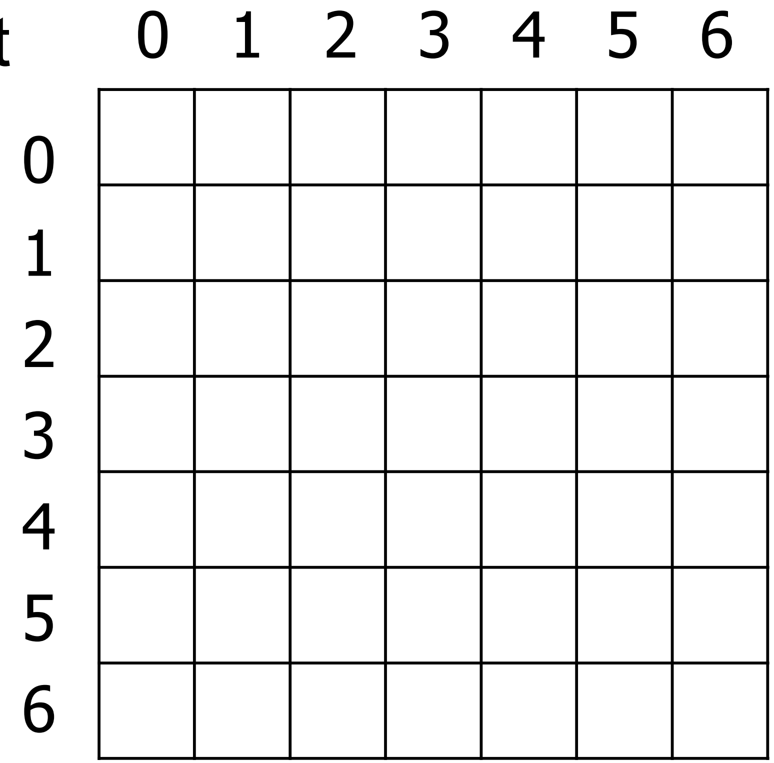
- Relevant reading: Chapters 1 and 2

Primitive Shapes

Point **Line** **Rectangle** **Ellipse**

Coordinates

- **The screen is like a piece of graph paper**
  - Each cell is a *pixel* ("picture element")
- **The *origin* (0, 0) is at the *top left***
  - x-axis: + is to the right, - is to the left
  - y-axis: + is down, - is up

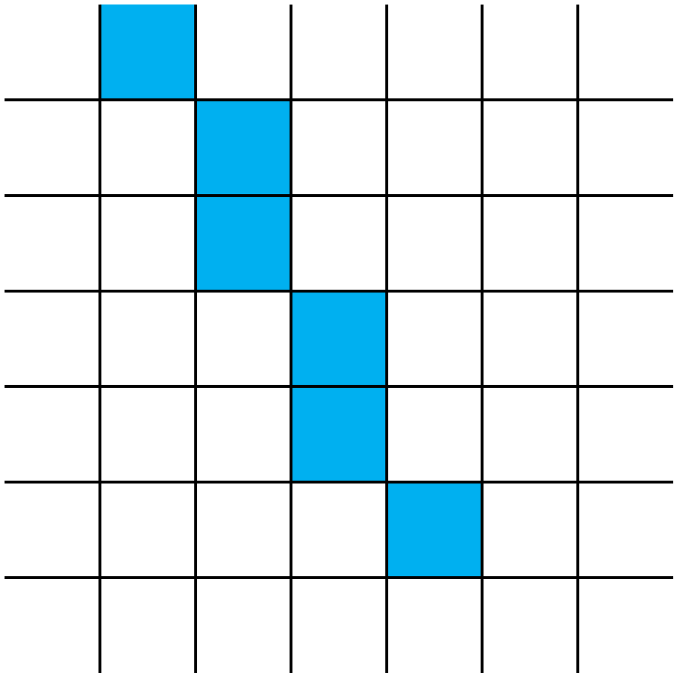


Points

- **Use the point() command to draw a single pixel**
- **point() takes two integers as input values:**
  - The x and y coordinates of the pixel
- **point() draws a point at the indicated location in the current drawing color**
- **Use a semicolon to end the command**
  - e.g., point(3, 4); draws a point at x: 3 and y: 4

Drawing Lines

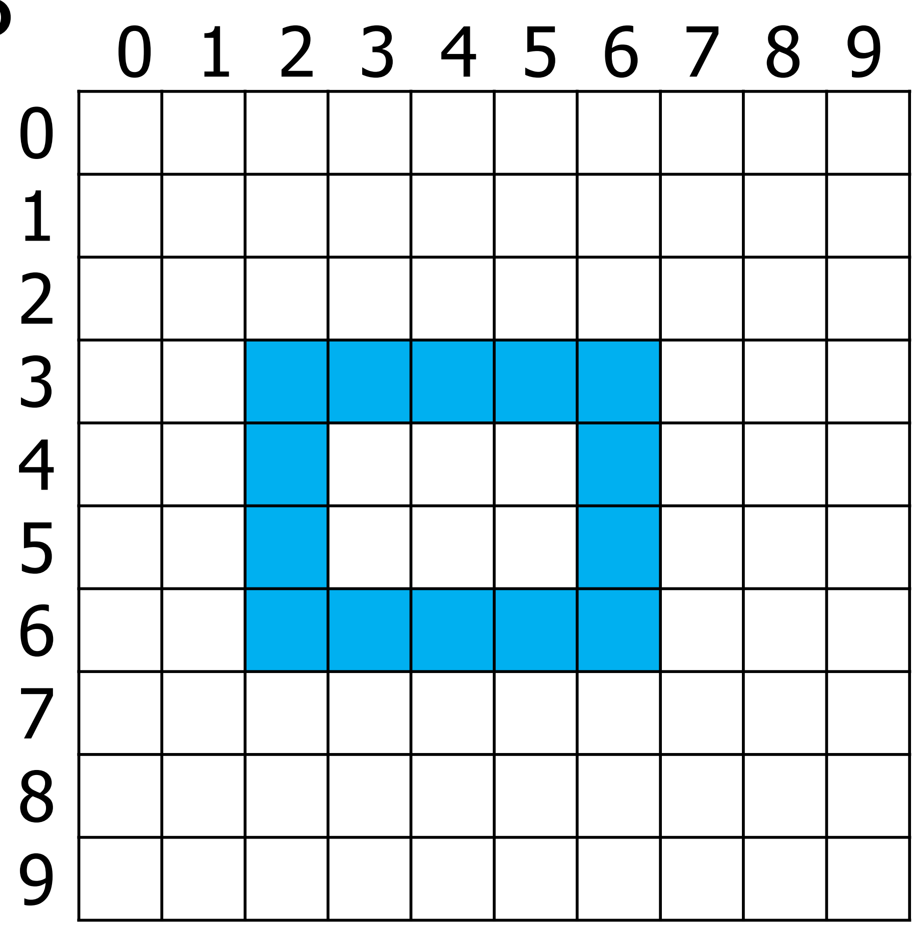
- To draw a line, use the line() command
- Tell line() where the line should begin and end
- Format:
  - line(*start_x, start_y, end_x, end_y*);



line(1,0,4,5);

Rectangles

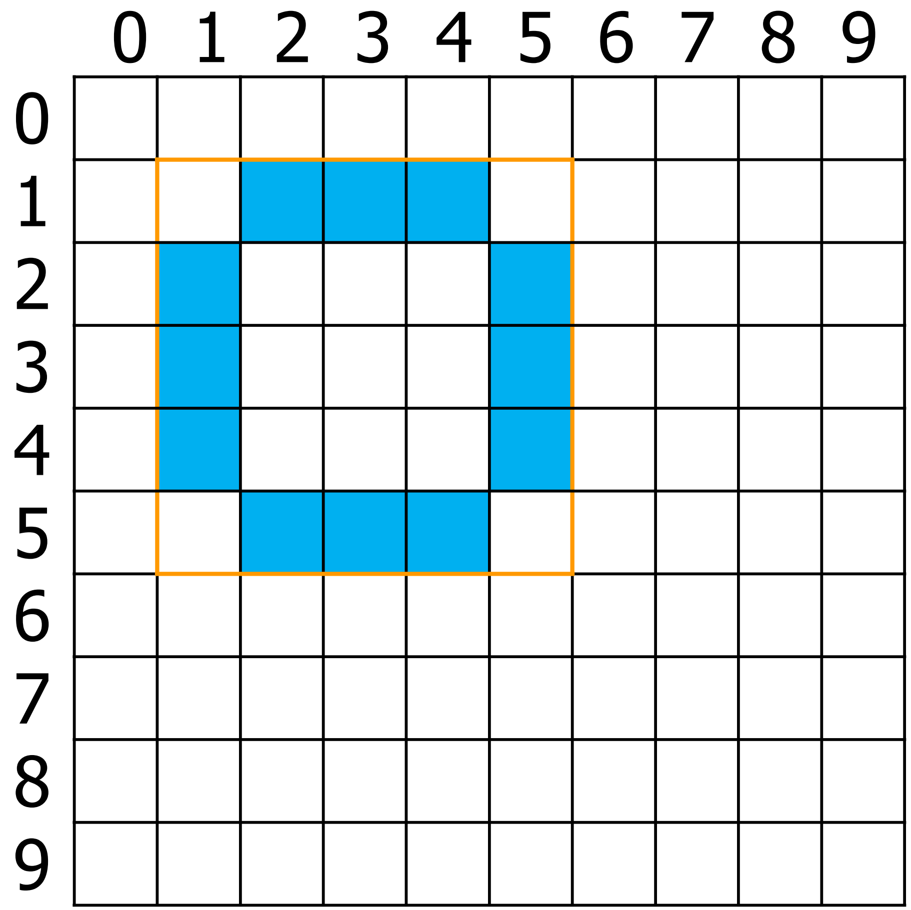
- **To draw a rectangle, you need to know:**
  - 1. the coordinates of its top-left corner
  - 2. its width (in pixels)
  - 3. its height (in pixels)



rect (2, 3, 5, 4);

Ellipses

- **Ellipses are drawn in the center of a bounding box (a rectangle that encloses all of the ellipse's points)**
- **To draw an ellipse, you need to know:**
  - 1. its center coordinates
  - 2. its width in pixels
  - 3. its height in pixels



ellipse (3, 3, 5, 5);

Triangles

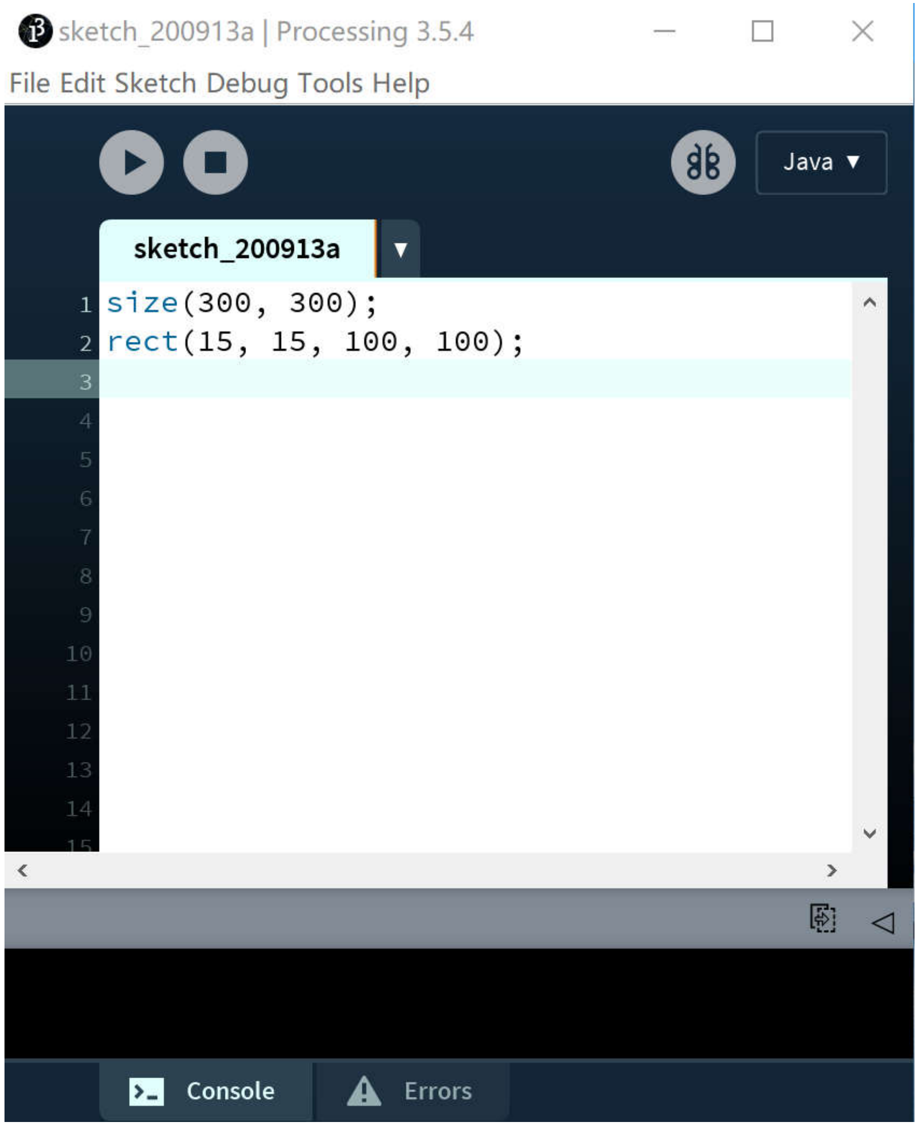
- Use the triangle() function to draw a triangle on the screen
- triangle() takes six values: the x and y coordinates of each vertex
  - ex. triangle (100, 50, 75, 100, 125, 100);

Canvas Size

- By default, Processing only gives you a small amount of drawing space (known as the “canvas”)
- The size() command tells Processing how much drawing space you want to use
  - size (*width_in_pixels* , *height_in_pixels*);
- You must set the canvas size **before** you draw anything!
  - This will normally be the first Processing command in any program you write

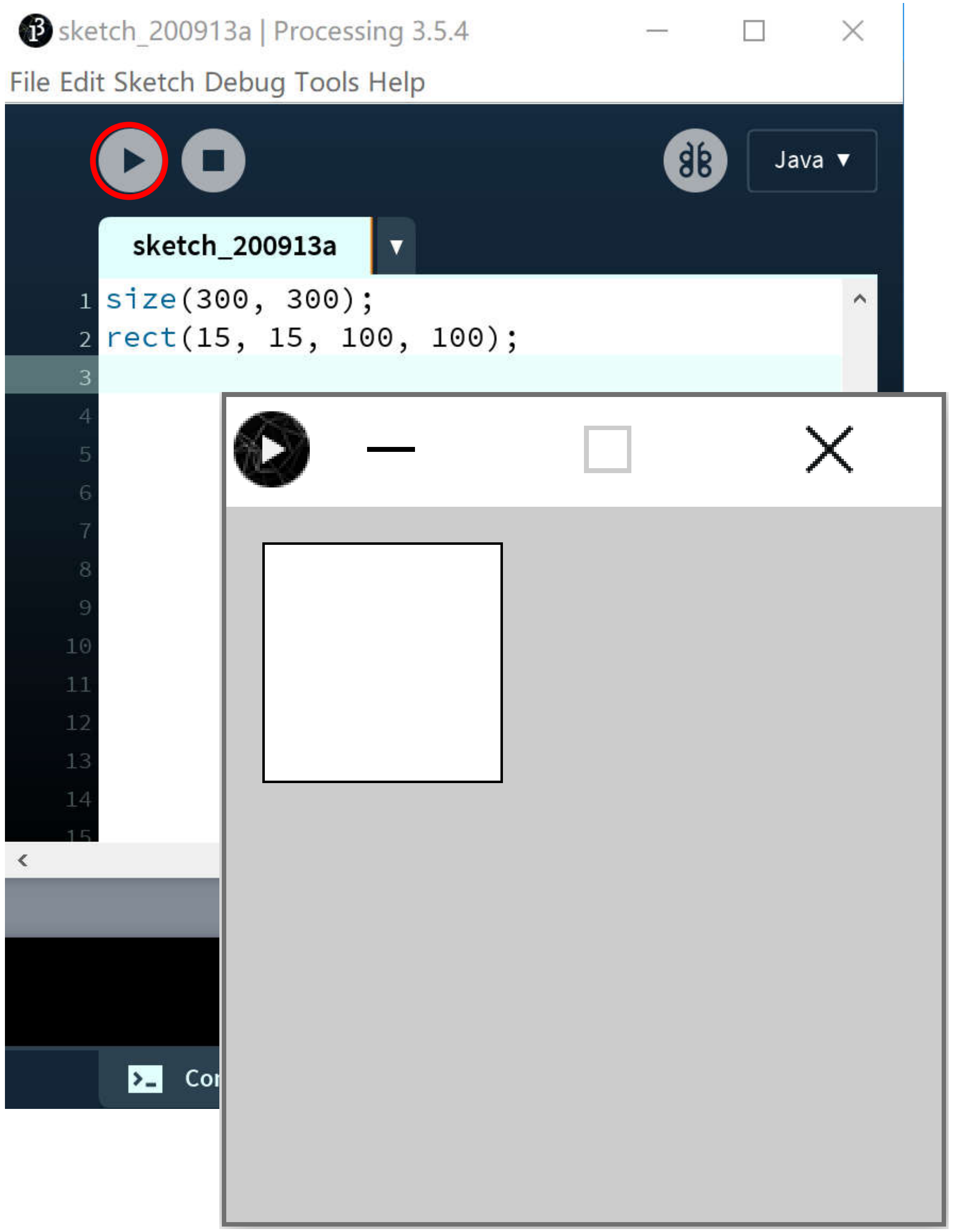
The Processing Sketchpad

- **Use the Sketchpad to write Processing code**



The Processing Sketchpad

- To run your code, click the **Run** button
- A new window will display the result



Adding Color

- **By default, Processing draws black lines and white objects on a gray background**
- **We can change the color of the background (canvas)**
- **We can also change the drawing (pen) color, and the fill color**

Color Types

- Color comes in two types: grayscale and RGB
- Grayscale is indicated by an integer between 0 (pure black) and 255 (pure white)
- RGB (Red-Green-Blue) uses three integers between 0 and 255 (one for each color component)
  - e.g., 255 red, 125 green, 8 blue = pumpkin orange
- You can also use Processing's Color Selector (in the Tools menu)

Grayscale and RGB

Grayscale Value		Color
40	=	
80	=	
120	=	
160	=	
200	=	
240	=	

Red	Green	Blue		Color
30	30	30	=	
100	30	30	=	
50	200	50	=	
50	50	150	=	
120	100	75	=	
175	50	255	=	

Canvas Color

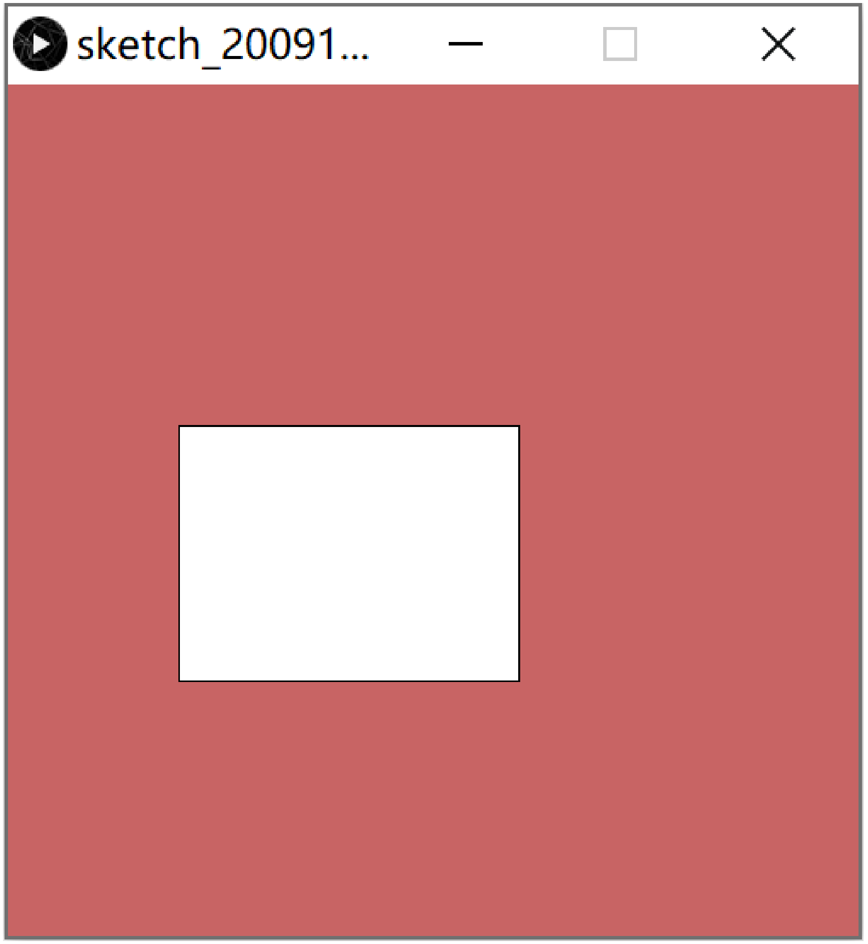
- Use the background() command to set the color of the canvas
- In the parentheses, put one integer for grayscale color or three (with commas) for RGB color
  - e.g., background (50);
  - e.g., background (255, 0, 255);

My First Program

```
size(500, 500);

background(200, 100, 100);

rect(100, 200, 200, 150);
```



Other Color Changes

- **stroke() changes line color**
  - **e.g., stroke(255, 0, 0);**
- **fill() changes the color inside a shape**
  - **e.g., fill(153);**

- Note 1: do this **BEFORE** drawing a shape!
- Note 2: Changing the stroke or fill color affects every line or shape you draw from that point on (at least, until the next color change)

Comments

- Comments are notes that describe how a program works
  - They are only useful for humans; the computer ignores them
- To add a comment, insert two consecutive forward slashes (//) in front of the comment text
- The comment extends from the slashes until the end of the line
  - e.g., // This is a comment